Austin Brown | C.V.

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Research interests

My main research focus is on the design, reliability, and computational complexity of Markov chain Monte Carlo algorithms by studying their convergence properties. My other research interests are in optimal transportation, weak convergence of stochastic processes, and error-in-variable models.

My dissertation "Some Convergence Results for Metropolis-Hastings Algorithms" is available online `https://conservancy.umn.edu/handle/11299/243073`.

Education

University of Minnesota - Twin Cities **Minneapolis, MN**
Ph.D. in Statistics *August 2022*
Dissertation: "Some Convergence Results for Metropolis-Hastings Algorithms"
Advisor: Galin L. Jones
Committee: Charles Geyer, Qian Qin, Wei-Kuo Chen

University of Florida **Gainesville, FL**
B.S. in Mathematics *July 2017*
Major in Mathematics and minor in Statistics

Appointments

University of Toronto - St George **Toronto, Ontario**
Postdoctoral Fellow *July 2023–Present*
Advisor: Jeffrey Rosenthal

University of Warwick **Coventry, England**
Research Fellow *September 2022–June 2023*
Advisors: Krzysztof Łatuszynski and Gareth Roberts

Publications and preprints

[1] A. Brown and J. S. Rosenthal, "Esssentially sharp rates for adaptive Markov chain Monte Carlo," *In progress*, 2024.

[2] A. Brown, K. Łatuszyński, and G. Roberts, "Adaptive pseudo-marginal Metropolis-

Hastings," *In progress*, 2024.

[3] A. Brown and J. S. Rosenthal, "Weak convergence of adaptive Markov chain Monte Carlo," *minor revision requested, Journal of Applied Probability*, 2024.

[4] S. Sixta, J. S. Rosenthal, and A. Brown, "Bounding and estimating MCMC convergence rates using common random number simulations," *ArXiv preprint, submitted Stochastic Models*, 2024.

[5] A. Brown, "A non-asymptotic error analysis for parallel Monte Carlo estimation from many short Markov chains," *ArXiv preprint, submitted Statistics and Computing*, 2024.

[6] A. Brown and G. L. Jones, "Exact convergence analysis for Metropolis–Hastings independence samplers in Wasserstein distances," *Journal of Applied Probability*, vol. 61, no. 1, p. 33–54, 2024.

[7] A. Brown and G. L. Jones, "Lower bounds on the rate of convergence for accept-reject-based Markov chains in Wasserstein and total variation distances," *Bernoulli, to appear*, 2024.

[8] A. Brown, "Geometric ergodicity of Gibbs samplers for Bayesian error-in-variable regression," *Electronic Journal of Statistics*, vol. 18, no. 1, pp. 1495–1516, 2024.

[9] A. Brown and G. L. Jones, "Convergence rates of Metropolis–Hastings algorithms," *WIREs Computational Statistics*, vol. 16, no. 5, p. e70002, 2024.

Seminar talks and conference posters

University of Warwick **Coventry, United Kingdom**
Algorithms and Computationally Intensive Inference Seminar *January 2023*
Talk Title: Lower Bounds on the Rate of Convergence for Accept-Reject-Based Markov Chains
Slides: `https://austindavidbrown.github.io/talk/warwick2023.pdf`

University of Warwick **Wales, United Kingdom**
University of Warwick Departmental Conference 2023 *April 2023*
Talk Title: Exact convergence for independence samplers in Wasserstein distance.
Slides: `https://austindavidbrown.github.io/talk/warwickdepartmental2023.pdf`

University of Oxford **England, United Kingdom**
Bioinference 2023 *June 2023*
Poster Title: Geometric ergodicity of Gibbs samplers for Bayesian error-in-variable regression
Poster: `https://austindavidbrown.github.io/poster/bioinference2023.pdf`

University of Florida **Gainesville, FL**
University of Florida Statistics Seminar *September 2023*
Talk Title: Lower Bounds for Metropolis-Hastings in Wasserstein distances
Slides: `https://austindavidbrown.github.io/talk/uf2023.pdf`

University of Toronto **Toronto, Canada**
University of Toronto B.B. Seminar *January 2024*
Talk Title: Error analysis for parallel Monte Carlo estimation from many short Markov chains
Slides: `https://austindavidbrown.github.io/talk/toronto2024.pdf`

Teaching experience

University of Minnesota - Twin Cities **Minneapolis, MN**
Instructor ($\sim$ 70 students) *Spring 2020*
Regression and Correlated Data (STAT 3032)

University of Minnesota - Twin Cities **Minneapolis, MN**
Instructor ($\sim$ 80 students) *Spring 2021*
Introduction to Statistical Analysis (STAT 3011)

University of Toronto - St. George **Toronto, Canada**
Instructor ($\sim$ 250 students) *Fall 2023*
Methods of Data Analysis 1 (STAT 302)

University of Toronto - St. George **Toronto, Canada**
Instructor ($\sim$ 300 students) *Fall 2024*
Methods of Data Analysis 1 (STAT 302)

Academic service

I have referred 6 articles for statistics and applied probability journals including Bayesian Analysis, Journal of Computational and Graphical Statistics, PLOS ONE, Advances in Applied Probability, Journal of Applied Probability, and Sankhya, Series B. I have also edited *An Introduction to Envelopes* by Dennis Cook.

I have taken part in the following committees and organizations:

American Statistical Association LGBTQ+ Advocacy Committee *August 2023–Present*

Organizer of University of Warwick Algorithms and Computational Intensive Seminar *Jan 2023–June 2023*

University of Warwick Statistics Department IT Committee *2022–June 2023*

University of Warwick Stonewall Self-Assessment Team (LGBTQ+ Inclusion) *2022–June 2023*

Organizations and memberships

Institute of Mathematical Statistics *2022–Present*

Funding, scholarships, and awards

Beverly and Richard Fink Fellowship ($56000+Tuition) *2017–2022*

Lynn Lin Fellowship ($2000) *2019–2019*

First Year Scholarship ($2500) *2017–2017*

Mentorships

University of Toronto - St George **Toronto, Canada**

2024

I am currently assisting in advising and mentoring 2 PhD students and 1 undergraduate student on their own individual research projects

Extracurricular activities

- Markov Chain Monte Carlo Group (2019–2022)
  Discussion of the state of the art MCMC literature

Statistical software

cmhi: A Python package for the centered Metropolis-Hastings independence algorithm. github.com/austindavidbrown/Centered-Metropolis-Hastings

mhlb: A Python implementation to estimate lower bounds on the geometric convergence rate for RWM Metropolis-Hastings. https://github.com/austindavidbrown/lower-bounds-for-Metropolis-Hastings

Programming language proficiency

Python: Pytorch, Autograd, Numpy, Matplotlib *Expert*

R: Rmarkdown *Expert*

C++: C++14 specification and above *Intermediate*

SQL: *Advanced*

Languages

- English (native)
- French (novice)
- Spanish (novice)

Professional experience

University of Minnesota - Twin Cities **Minneapolis, MN**

Graduate Assistant/PhD Candidate *2017–2022*

Projects and responsibilities:

- As a graduate instructor and teaching assistant, I developed course materials, taught lectures, and administered exams for undergraduate and graduate courses.
- Project title: l-net: Extending the elastic-net with l1 to l10 penalties in regression.
  - Abstract: We create an R package written in C++ to combine l^1 penalized regression with higher order l^p penalizations built upon the Elastic-net. The package is able to fit regression models with penalizations ranging from l^1 to l^{10}.
  - Link: github.com/austindavidbrown/l-net
- Project title: Confidence Intervals via Gaussian Policy Gradients as an Augmented Wasserstein Gradient Flow.
  - Abstract: We introduce a novel reinforcement learning model by reformulating the policy-gradient algorithm as an augmented Wasserstein gradient flow. This formulation leads to statistically valid confidence intervals when using Gaussian policies. We focus our applications on obtaining confidence intervals in supervised learning. In image classification on CIFAR10, our model performs comparable to existing approaches and on MNIST, our model outperforms existing reinforcement learning approaches.

University of Minnesota - Twin Cities **Minneapolis, MN**

Research Assistant *2020–2020*

Project description: Applied Bayesian inference to deep learning for high-dimensional protein structure prediction.

- Principal Investigator: Singdhansu Chatterjee, NSF award: 1939956

University of Minnesota - Twin Cities **Minneapolis, MN**

Statistical Consultant *2019–2019*

Statistical consultant for the Initiative Institute on the Environment at the University of Minnesota.

- Project title: Modeling Corn Yield by Temperature and Precipitation
- Description: Extreme temperature and precipitation prediction using deep Gaussian processes for the state of Minnesota.

University of Florida **Gainesville, FL**

Full Time Student *2017*

Lightmaker **Orlando, FL**

Lead Mobile Software Engineer/Mobile Software Engineer *2012–2015*

I engineered awesome mobile applications and assisted with optimization and performance of mobile websites.

- Technologies used: Java, C, iOS, Java, Git.
- Notable projects:
  - Barclays ATP World Tour Finals;
  - Mayo Clinic;
  - AMC Extras iOS;
  - LPGA Android/iOS application;
  - Mcafee Partner iOS;
  - USGA Rules of Golf iOS/ Android;
  - Cherokee Casinos;
  - Hard Rock Tulsa;
  - Ikea;

References

- Professor Galin Jones. University of Minnesota - Twin Cities. Email: galin@umn.edu
- Assistant Professor Qian Qin. University of Minnesota - Twin Cities. Email: qqin@umn.edu
- Associate Professor Krzysztof Latuszynski. University of Warwick. Email: K.G.Latuszynski@warwick.ac.uk